```
!pip -q install openpyxl

import pandas as pd

df = pd.read_excel(
    "/content/Copy of Muskoka_Area_Intensive_Sentinel lakes_Temperature_and_dissolved_oxygen_1976-2021.xlsx",
    engine="openpyxl"
)

display(df.head())
```

	LAKE	DISTRICT	TOWNSHIP	LATITUDE	LONGITUDE	DATE	YEAR	MONTH	SEASON	DEPTH	TEMPERATURE	DISSOLVED OXYGEN	STATION
0	BLUE CHALK LAKE	Muskoka Dist.	Ridout	451156	785631	1976-02-18	1976	2	Winter	0.1	0.90	NaN	3007754001
1	BLUE CHALK LAKE	Muskoka Dist.	Ridout	451156	785631	1976-02-18	1976	2	Winter	1.0	2.40	NaN	3007754001
2	BLUE CHALK LAKE	Muskoka Dist.	Ridout	451156	785631	1976-02-18	1976	2	Winter	2.0	3.00	NaN	3007754001
3	BLUE CHALK	Muskoka Dist	Ridout	451156	785631	1976-02-18	1976	2	Winter	3.0	3.75	NaN	3007754001

```
df.shape
```

```
(104319, 13)
```

```
df.isnull().sum()
```

	0
LAKE	0
DISTRICT	0
TOWNSHIP	0
LATITUDE	0
LONGITUDE	0
DATE	0
YEAR	0
MONTH	0
SEASON	0
DEPTH	0
TEMPERATURE	2889
DISSOLVED OXYGEN	26763
STATION	0

dtype: int64

```
blank_percentage = (df.isnull().sum() / len(df)) * 100

print('Percent of missing variables:',blank_percentage.round(2))
```

```
Percent of missing variables: LAKE                0.00
DISTRICT            0.00
TOWNSHIP            0.00
LATITUDE            0.00
LONGITUDE           0.00
DATE                0.00
```

```
YEAR                0.00
MONTH               0.00
SEASON              0.00
DEPTH               0.00
TEMPERATURE         2.77
DISSOLVED OXYGEN   25.65
STATION             0.00
dtype: float64
```

```
df[["DEPTH", "TEMPERATURE", "DISSOLVED OXYGEN",""]].describe()
```

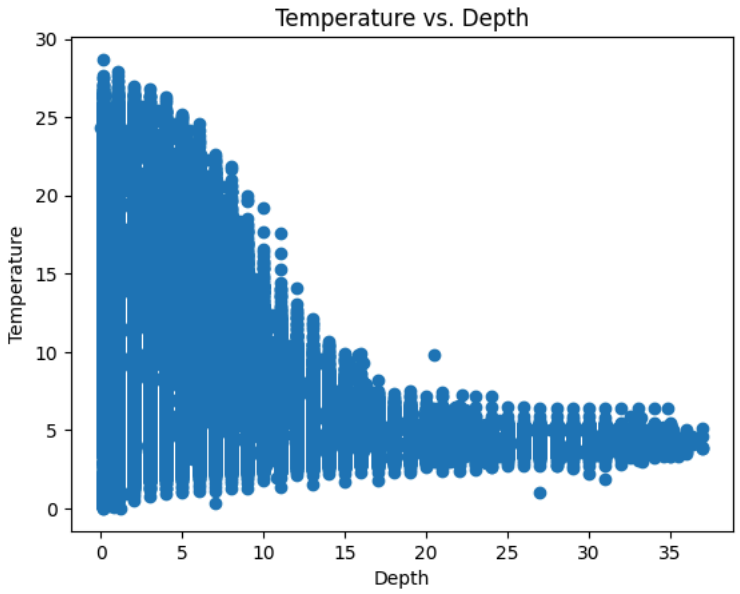
	DEPTH	TEMPERATURE	DISSOLVED OXYGEN	MONTH
count	104319.000000	101430.000000	77556.000000	104319.000000
mean	11.564774	8.514137	7.049976	7.195774
std	8.684943	5.915419	3.239833	2.524883
min	0.000000	0.000000	0.000000	1.000000
25%	4.500000	4.500000	5.070000	5.000000
50%	10.000000	5.630000	7.890000	7.000000
75%	17.000000	10.800000	9.300000	9.000000
max	37.000000	28.700000	17.000000	12.000000

```
import matplotlib.pyplot as plt
temperature_depth = df.dropna(subset=['TEMPERATURE', 'DEPTH'])

plt.scatter(temperature_depth['DEPTH'], temperature_depth['TEMPERATURE'])
plt.xlabel('Depth')
plt.ylabel('Temperature')
plt.title('Temperature vs. Depth')

print("Correlation between temperature and depth:",temperature_depth['TEMPERATURE'].corr(temperature_depth['DEPTH']))
```

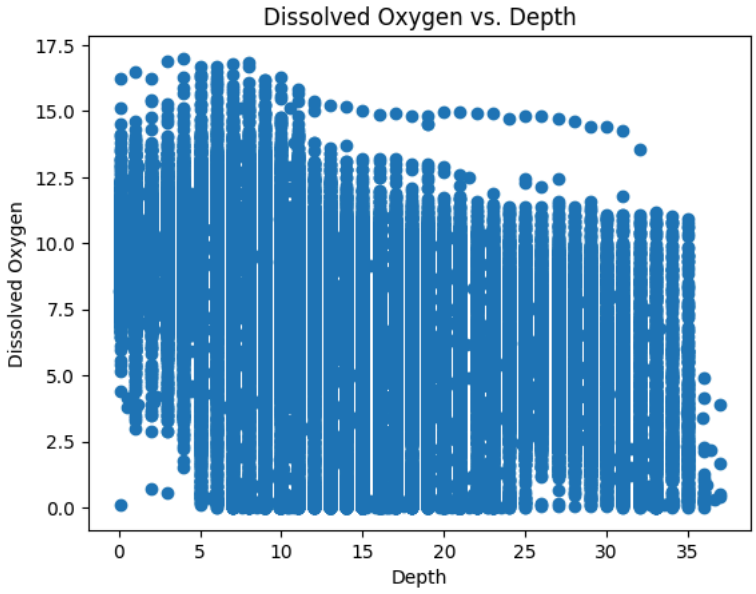
```
Correlation between temperature and depth: -0.5946528977358294
```




```
depth_dissolved_oxygen = df.dropna(subset=['DISSOLVED OXYGEN', 'DEPTH'])
plt.scatter(depth_dissolved_oxygen['DEPTH'], depth_dissolved_oxygen['DISSOLVED OXYGEN'])
plt.xlabel('Depth')
plt.ylabel('Dissolved Oxygen')
plt.title('Dissolved Oxygen vs. Depth')

print("Correlation between dissolved oxygen and depth:",depth_dissolved_oxygen['DEPTH'].corr(depth_dissolved_oxygen['DISSOLVED
```

```
Correlation between dissolved oxygen and depth: -0.46855511303863245
```




```
Start coding or generate with AI.
```

```
temperature_dissolved_oxygen = df.dropna(subset=['DISSOLVED OXYGEN', 'TEMPERATURE'])
plt.scatter(temperature_dissolved_oxygen['TEMPERATURE'], temperature_dissolved_oxygen['DISSOLVED OXYGEN'])
plt.xlabel('Temperature')
plt.ylabel('Dissolved Oxygen')
plt.title('Dissolved Oxygen vs. Temperature')

print("Correlation between temperature and dissolved oxygen:",temperature_dissolved_oxygen['TEMPERATURE'].corr(temperature_disso
```

```
Correlation between temperature and dissolved oxygen: 0.2558877526098963
```

